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Rhein and Aloe-Emodin Kinetics from Senna Laxatives in Man

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Abstract
Therapeutic doses of two laxatives (Agiolax® and Sennatin®) were repeatedly administered to 10 healthy volunteers in a two-way change-over design. Blood samples were collected up to 96 h after the first dose, and plasma levels of total aloe-emodin and rhein were determined simultaneously with a sensitive (lower limit of quantification: 0.5 ng aloe-emodin and 2.5 ng rhein per millilitre plasma) and specific fluorometric HPLC method. Aloe-emodin was not detectable in any plasma sample of any subject. Rhein concentration time courses showed highest levels of 150–160 ng/ml and peak maxima at 3–5 h and 10–11 h after dosing probably according to absorption of free rhein and rhein released from prodrugs (e.g. sennosides) by bacterial metabolism, respectively.

Introduction

Senna extracts from senna leaves or pods or finely ground plant material are widely used as laxatives. Their main constituents are sennosides which are transformed by bacterial enzymes to sennidins and anthrones with subsequent oxidation to anthraquinones outside the reductive atmosphere of the large intestine [1, 2]. The class of sennosides, though on the whole 10,10′-bisanthronyl compounds, is not uniform in chemical structure. Sennosides A and B are homodianthrone diglucosides of rhein (COOH substituent) and sennosides C and D, being present in much smaller amounts, are heterodianthrone diglucosides bearing a –COOH and a $-CH_2OH$ substituent. The latter ones, after metabolic

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decomposition, may give rise to levels of aloe-emodin (1,8-dihydroxy-3-hydroxymethylanthraquinone) in blood. As aloe-emodin has been reported to possess genotoxic activity in certain in vitro incubation tests [3, 4], it was the purpose of this investigation to quantify aloe-emodin absorption after single and repeated administration of two laxatives, Agiolax® granulate (from dried and ground senna pod) and Sennatin® (purified sennosides in natural isomer mixture), both manufactured by Madaus AG, Cologne, FRG, for estimating the toxicological relevance of this compound.

Experimentation

Material

Rhein (1,8-dihydroxyanthraquinone-3-carboxylic acid) and danthron (1,8-dihydroxyanthraquinone) were purchased from Roth, Karlsruhe, FRG, and Aldrich, Beerse, Belgium in the highest purity available. Aloe-emodin (1,8-dihydroxy-3-hydroxymethylanthraquinone) was supplied by the Chemical Research Department of Madaus AG. All solvents (acetonitrile, di-isopropylether, glacial acetic acid and methanol) were of analytical grade or Lichrosolv® quality and were obtained from Merck, Darmstadt, FRG. Water was double distilled in our laboratories before use. Sodium acetate and concentrated (37%) hydrochloric acid, which was diluted to approximately 6 mol/l before use, were supplied from Merck and sulphatase (containing β-glucuronidase activity, catalogue No. S-9751) from Sigma, Deisenhofen, FRG. Blank human plasma, used for calibration samples, was made available by the blood bank of the Municipal Hospital Köln-Merheim, FRG.

To prepare 1 mol/l sodium acetate buffer, 82.03 g sodium acetate (anhydrous) were dissolved in 1 litre of water, and the pH was adjusted to 5.6 by dropwise addition of glacial acetic acid.

HPLC (Solvents and Instrumentation)

HPLC separations were carried out on a Nucleosil® 120–3 C_{18} reversed-phase column (125 mm × 3 mm) with a precolumn (11 mm × 3 mm) filled with the same material. The mobile phase was acetonitrile/methanol/acetic acid (1 mol/l) mixed in a volume ratio of 33:7:60, which was delivered by a Series 4 Liquid Chromatograph (Perkin-Elmer, Norwalk, Conn., USA) with a flow rate of 0.4 ml/min at 40 ± 1 °C. Samples were injected with an ISS-100 Sampling System (Perkin-Elmer). Detection was made with a LC-3000 Fluorescence Spectrometer (Perkin- Elmer) at an emission wavelength of λ_{em} = 530 nm (excitation wavelength: λ_{ex} = 430 nm). After resolution, peak heights were calculated by P-E Nelson Turbochrom 3 (version 3.3) data acquisition and integration software installed on a 80486 Personal Computer (50 MHz). Retention times of aloe-emodin, rhein and danthron were 7.3, 8.5 and 18.0 min, respectively.

Sample Preparation

1 ml plasma was made up with 1 mol/l sodium acetate buffer (pH 5.6) to a final volume of 2.5 ml, internal standard (20 ng danthron/50 μl methanol) and sulphatase (50 μl, undiluted) were added and the sample was hydrolysed under gentle shaking at 37 °C overnight. After that, the sample was acidified with 600 μl of 6 mol/l hydrochloric acid to a pH of $\leqslant$ 3.5, extracted for 20 min with 6 ml di-isopropylether and centrifuged at 2,000 *g* for phase separation. 5 ml of the organic phase were aspirated, transferred to a 5-ml Reacti-Vial® (Pierce) and evaporated to dryness at 37 °C under a steam of nitrogen. The solid residue was reconstituted in 30 μl methanol, from which 10 μl were injected onto the column.

Calibration

For calibration, 1 ml human blank plasma was spiked with 250–2.5 ng rhein and 25–0.25 ng aloe-emodin in a constant concentration ratio of 10:1, dissolved in common in 50 μl methanol. The samples were diluted with 1 mol/l sodium acetate buffer (pH 5.6) to 2.5 ml and, after addition of internal standard (20 ng danthron/50 μl methanol), were hydrolysed with 50 μl sulphatase overnight. Acidification, extraction and all further procedures were as described in the previous section.

Calculation

Peak height ratios of rhein or aloe-emodin and the internal standard danthron are used for calculation. There is a linear relationship of relative fluorescence intensity to concentration of rhein and aloe-emodin in the concentration range examined. The lower limit of determination is 2.5 ng rhein and 0.5 ng aloe-emodin per millilitre as judged by a non-precision and non-accuracy of less than 10% at day-to-day recovery measurements.

Treatment of Subjects

The clinical part of the pharmacokinetic study was carried out with a group of 10 male healthy volunteers (age 18–32 years, average age 26.8 years) at the Laboratory for Contract Research in Clinical Pharmacology and Biopharmaceutical Analytics GmbH (Bad Schwartau, FRG). Each subject received in a two-way change-over design 4 single doses of Agiolax (6.3 g granulate containing 13.23 mg total anthranoids including 7.62 mg potential rhein and 378 μg potential aloe-emodin) or Sennatin (2 tablets containing 20.36 mg total anthranoids including 13.06 mg potential rhein and 400 μg potential aloe-emodin) at 24-hour intervals. 'Potential' means the sum of rhein or aloe-emodin being present as free anthraquinone and in the form of the corresponding dianthrones (sennosides, sennidins) and monoanthrone glucosides. On days 1 and 4, blood samples of 5 ml were taken according to the following sampling scheme: 0 (prior to dose) and 30 min, 1 h and every hour during 2 and 16 h after dose, then every second hour until 24 h after dose. On days 2 and 3, blood was taken only immediately before treatment. Blood samples were collected in heparinized tubes, centrifuged at 1,600 *g* and the plasma supernatants were aspirated and deep frozen at −20 °C until analysis.

Results

Aloe-Emodin (1,8-Dihydroxy-3-Hydroxymethylanthraquinone)

Aloe-emodin, administered in different binding states (mainly sennosides C and D, much less aloe-emodin glucoside and free aloe-emodin) with one single dose of Agiolax and Sennatin in quantities of 378 and 400 μg could not be detected in any blood sample of any volunteer throughout the period of observation, thus its concentration is less than 0.5 ng/ml. There was absolutely no evidence for accumulation of aloe-emodin following repeated (fourfold) intake of Agiolax or Sennatin in therapeutic doses which is in full accordance with former findings from a pilot study with 6 volunteers who received 7 equal doses of Agiolax or Sennatin in a parallel design (data not shown).

Rhein (1,8-Dihydroxyanthraquinone-3-Carboxylic Acid)

Pharmacokinetic parameters derived from individual total rhein plasma levels are summarized in tables 1 and 2 with mean values and standard deviations for all subjects. Single concentration-time data are not presented.

The data show that following Sennatin (table 1) and Agiolax (table 2) intake total rhein concentration levels in plasma do not exceed 160 ng/ml during the whole period of observation. Including the last time point (96 h after first treatment) rhein could be measured with sufficient precision and accuracy according to our criteria of reliability.

On the average, the highest concentration values of total rhein occurred 10–11 h after the first or fourth dose of Sennatin corresponding to the time needed to transport sennosides to the large intestine and to metabolize them there; preceding relative maxima perceptible 1–3 h after administration are probably due to absorption of unbound rhein in the formulation.

Maximal concentrations of total rhein following Agiolax administration appear much earlier (mean: 3–5 h after treatment) indicating absorption of higher amounts of free rhein in the drug. A subsequent increase in rhein plasma levels (caused by release of rhein from sennosides) is less significant.

Discussion

Following repeated administration of Agiolax or Sennatin, aloe-emodin and rhein were determined in plasma samples of healthy male volunteers with a validated, rugged, highly sensitive and specific fluorometric HPLC method.

Rhein concentration time courses followed the expected profile with concentration maxi-

Table 1. Pharmacokinetic behaviour of total rhein following repeated Sennatin® administration to man

	Volunteer No.										Mean
	1	2	3	4	5	6	7	8	9	10	
t_{max}, h											
1st day	16	11	13	11	12	14	14	8	12	2	11.3 (3.9)
4th day	13	11	10	12	12	7	13	1	9	9	9.7 (3.6)
C_{max}, ng/ml											
1st day	37.0	44.9	28.6	23.6	43.6	74.5	40.2	33.7	45.2	66.3	43.8 (15.8)
4th day	30.1	39.8	47.7	30.9	44.8	65.4	66.9	66.9	48.0	55.4	49.6 (13.9)
$t_{½}$, h											
4th day (terminal)	12.1	7.1	6.2	8.0	4.5	6.7	9.1	5.8	5.4	4.7	7.0 (2.3)

SD are indicated in parentheses.

Table 2. Pharmacokinetic behaviour of total rhein following repeated Agiolax® administration to man

	Volunteer No.										Mean
	1	2	3	4	5	6	7	8	9	10	
t_{max}, h											
1st day	3	3	2	4	4	9	3	3	2	1	3.4 (2.2)
4th day	5	9	9	4	0.5	2	7	8	2	2	4.9 (3.2)
C_{max}, ng/ml											
1st day	53.7	85.5	59.9	55.6	70.1	68.3	69.5	67.0	60.7	61.1	65.1 (9.2)
4th day	75.0	63.7	49.8	103.4	43.8	135.4	156.3	53.1	71.0	66.4	81.8 (37.9)
$t_{½}$, h											
4th day (terminal)	10.8	7.3	7.5	5.0	4.0	6.7	9.3	7.6	6.8	8.0	7.3 (1.9)

SD are indicated in parentheses.

ma almost immediately after dosing (absorption of free rhein) and approximately 10 h later (absorption of rhein liberated from sennosides and sennidins by reductive bacterial cleavage in the large intestine and subsequent oxidation).

Aloe-emodin, though co-administered in measurable amounts, was not detectable in any plasma sample of any subject within the duration of the study. With regard to a lower limit of quantification of 0.5 ng aloe-emodin/ml and a lower limit of reliable identification of approximately 0.2 ng aloe-emodin/ml plasma, concentration levels, if at all existing, are lower by a factor of $\geq 20{,}000$ than concentrations with potential in vitro genotoxic properties. Consequently, treatment of constipation with Agiolax or Sennatin according to dose recommendations is considered to involve no genotoxic risk in man.

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